

Last login: Tue Jun 30 16:14:09 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb
psql (18.4)
Type "help" for help.

tpch-200mb=# SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
  s_name,
  s_address
from
  supplier_secure,
  nation
where
  s_suppkey in (
    select
      ps_suppkey
    from
      partsupp_secure
    where
      ps_partkey in (
        select
          p_partkey
        from
          part
        where
          p_name like '%ivory%'
      )
    and ps_availqty > (
      select
        0.5 * sum(l_quantity)
      from
        lineitem_secure
      where
        l_partkey = ps_partkey
        and l_suppkey = ps_suppkey
        and l_shipdate >= date '1995-01-01'
        and l_shipdate < date '1995-01-01' + interval '1' year
    )
  )
  and s_nationkey = n_nationkey
  and n_name = 'FRANCE'
order by
  s_name;
```

SET

WARNING: Log 1: Planner Input:

DETAIL: {QUERY

```
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
  :alias <>
  :eref
    {ALIAS
      :aliasname supplier_secure
      :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
    }
  }
):rtekind 1
:subquery
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks true
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
      {RANGETBLENTY
      :alias
      :alias
        {ALIAS
          :aliasname s
          :colnames <>
        }
      :eref
        {ALIAS
          :aliasname s
          :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_ph
one" "s_acctbal" "s_comment")
        }
      :rtekind 0
      :relid 17399
      :inh true
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCI true
      :securityQuals <>
    )
  }
):rteperminfos (
  {RTEPERMISSIONINFO
  :relid 17399
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 9 10 11 12 13 14)
  :insertedCols (b)
  :updatedCols (b)
  }
)
```

:jointree

{FROMEXPR

:fromlist (

{RANGETBLREF

:rtindex 1

)

:quals

{BOOLEXP

:boolop and

:args (

{OPEXP

:opno 1756

:opfuncid 1720

:opresulttype 16

:opretset false

:opcollid 0

:inputcollid 0

```

:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {FUNCEXP
    :funcid 1740
    :funcresulttype 1700
    :funcresultset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:location -1
)
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBENTRY
        :alias
          {ALIAS
            :aliasname n
            :colnames <>
          }
        :eref
          {ALIAS
            :aliasname n
            :colnames ("n_nationkey" "n_name" "n_regionkey" "n_
              comment")
          }
        :rtekind 0
        :relid 17376
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
      {RANGETBENTRY
        :alias
          {ALIAS
            :aliasname r
            :colnames <>
          }
        :eref
          {ALIAS
            :aliasname r
            :colnames ("r_regionkey" "r_name" "r_comment")
          }
        :rtekind 0
        :relid 17382
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 2
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
    )
  }
:rtperminfos (
  {RTEPERMISSIONINFO
    :relid 17376
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17382
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
      {RANGETBLREF
        :rtindex 2
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {OEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
  }
  :resno 1
  :resname ?column?
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>

```

```

        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supplekey
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varstype 1042
:varstypepm 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varstype 1043
:varstypepm 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varstype 1042
:varstypepm 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 1700
:varstypepm 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 6
:resjunk false
}

```

```

{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 7
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 19006
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtskind 0
:relid 17376
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 19006
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:sublinkType 2
:subLinkId 0
:testexpr
{OPEXP
:opno 96
:opfncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 136
}
{PARAM
:paramkind 2
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
}
:location 146
}
)
:operName ("=")
:subselect
{QUERY
:commandType 1

```

```

:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp_secure
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_su
          pplycost" "ps_comment")
      }
    :rtekind 1
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTY
            :alias
              {ALIAS
                :aliasname ps
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname ps
                :colnames ("ps_partkey" "ps_suppkey" "ps_availqty"
                  "ps_supplycost" "ps_comment")
              }
            :rtekind 0
            :relid 17409
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromCl true
            :securityQuals <>
          }
        )
        :rteperminfos (
          {RTEPERMISSIONINFO
            :relid 17409
            :inh true
            :requiredPerms 2
            :checkAsUser 10
            :selectedCols (b 8 9 10 11 12)
            :insertedCols (b)
            :updatedCols (b)
          }
        )
      }
    :jointree
      {FROMEXPR
        :fromlist (
          {RANGETBLREF
            :rtindex 1
          }
        )
        :quals
          {BOOLEXP
            :boolop and
            :args (
              {OPEXP
                :opno 521
                :opfuncid 147
                :opresulttype 16
                :opretset false
                :opcollid 0
                :inputcollid 0
                :args (
                  {VAR
                    :varno 1
                    :varattno 3
                    :vartype 23
                    :vartypmod -1
                    :varcollid 0
                    :varnullingrels (b)
                    :varlevelsup 0
                    :varreturningtype 0
                    :varnosyn 1
                    :varattnosyn 3
                    :location -1
                  }
                  {CONST
                    :consttype 23
                    :consttypmod -1
                    :constcollid 0
                    :constlen 4
                    :constbyval true
                    :constisnull false
                    :location -1
                    :constvalue 4 [ 100 0 0 0 0 0 0 ]
                  }
                )
                :location -1
              }
            )
          }
        {SUBLINK
          :sublinkType 0
          :sublinkid 0
          :testexpr <>
          :operName <>
          :subselect
            {QUERY
              :commandType 1
              :querySource 0
              :canSetTag true
              :utilityStmt <>
              :resultRelation 0
              :hasAggs false
              :hasWindowFuncs false
              :hasTargetSRFs false
              :hasSubLinks false
              :hasDistinctOn false
              :hasRecursive false
              :hasModifyingCTE false
              :hasForUpdate false
              :hasRowSecurity false
              :hasGroupRTE false
            }
          }
        )
      }
    )
  }
)

```

```

:isReturn false
:ctelList <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_suppkey" "s_name" "s_addr
        ess" "s_nationkey" "s_phone" "s_acctbal
        " "s_comment")
    }
    :rtekind 0
    :relid 17399
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rtepermInfos (
  {RTEPERMISSIONINFO
    :relid 17399
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromList (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location -1
        }
        {VAR
          :varno 1
          :varattno 2
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 2
          :location -1
        }
      )
      :location -1
    }
    {OPEXP
      :opno 1756
      :opfuncid 1720
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 6
          :vartype 1700
          :vartypmod 983046
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        }
        {FUNCEXP
          :funcid 1740
          :funcresulttype 1700
          :funcset false
          :funcvariadic false
          :funcformat 2
          :funccollid 0
          :inputcollid 0
          :args (
            {CONST
              :consttype 23
              :consttypmod -1
              :constcollid 0
              :constlen 4
              :constbyval true
              :constisnull false
              :location -1
              :constvalue 4 [ 0 0 0 0 0 0 0
                0 ]
            }
          )
          :location -1
        }
      )
      :location -1
    }
  )
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
    }
  }
)

```

```

        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    )
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname ps_partkey
    :ressortgrouppref 0
    :resorigtbl 17409
    :resorigcol 1
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
    :resno 2
    :resname ps_supplekey
    :ressortgrouppref 0
    :resorigtbl 17409
    :resorigcol 2
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    :resno 3
    :resname ps_availqty
    :ressortgrouppref 0
    :resorigtbl 17409
    :resorigcol 3
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 4
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
    }
    :resno 4
    :resname ps_supplycost
    :ressortgrouppref 0
    :resorigtbl 17409
    :resorigcol 4
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 5
    :vartype 1043
    :vartypmod 203
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location -1
    }

```



```

    }
    :resno 5
    :resname ps_comment
    :ressortgroupref 0
    :resorigtbl 17409
    :resorigcol 5
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 19810
:inh false
:relkind v
:relockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 19810
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
  }
)
):jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
):quals
{BOOLEXP
  :boolop and
  :args (
    {SUBLINK
      :subLinkType 2
      :subLinkId 0
      :testexpr
      {OEXPR
        :opno 96
        :opFuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location 316
          }
          {PARAM
            :paramkind 2
            :paramid 1
            :paramtype 23
            :paramtypmod -1
            :paramcollid 0
            :location -1
          }
        )
      }
      :location 327
    }
  )
  :operName ("=")
  :subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
      {RANGETBENTRY
        :alias <>
        :eref
        {ALIAS
          :aliasname part
          :colnames ("p_partkey" "p_name" "p_mfgr" "p_b
            rand" "p_type" "p_size" "p_container" "p_reta
            ilprice" "p_comment")
        }
      }
      :rtkind 0
      :relid 17387
      :inh true
      :relkind r
      :relockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    )
  )
}
)rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17387
    :inh true
    :requiredPerms 2
    :checkAsUser 0
  }
)

```

```

        :selectedCols (b 8 9)
        :insertedCols (b)
        :updatedCols (b)
    }
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 1
:varattno 2
:vartype 1043
:vartypmod 59
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 581
}
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
)
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 593
:constvalue 11 [ 44 0 0 0 37 105 118 111 1
14 121 37 ]
}
)
:location 588
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 411
}
}
:resno 1
:resname p_partkey
:ressortgroupref 0
:resorigtbl 17387
:resorigcol 1
:resjunk false
}
)
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 327
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 657
}
}
)
}
:location -1
}
{SUBLINK
:sublinkType 4
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY

```

```

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem_secure
        :colnames ("l_orderkey" "l_partkey" "l_sup
pkey" "l_linenumber" "l_quantity" "l_exten
dedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitd
ate" "l_receiptdate" "l_shipinstruct" "l_s
hipmode" "l_comment")
      }
    :rtekind 1
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTY
            :alias
              {ALIAS
                :aliasname l
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname l
                :colnames ("l_orderkey" "l_partkey"
"l_suppkey" "l_linenumber" "l_quantiti
ty" "l_extendedprice" "l_discount" "
l_tax" "l_returnflag" "l_linestatus"
"l_shipdate" "l_commitdate" "l_rece
iptdate" "l_shipinstruct" "l_shipmod
e" "l_comment")
              }
            :rtekind 0
            :relid 17440
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesampl <>
            :lateral false
            :inFromCl true
            :securityQuals <>
          }
        )
      }
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 17440
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 9 10 11 12 13 14 15
16 17 18 19 20 21 22 23)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  )
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks false
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTY
            :alias
              {ALIAS
                :aliasname o
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname o
                :colnames ("o_orderkey"
"o_custkey" "o_orderstat
us" "o_totalprice" "o_or
derdate" "o_orderpriorit
y" "o_clerk" "o_shipprio
rity" "o_comment")
              }
            :rtekind 0
            :relid 17428
            :inh true
            :relkind r

```

```

:rellockmode 1
:perminfoindex 1
:tablesampl <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17428
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
){VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 1
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
){SCALARARRAYOPEXP
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 6
:location -1
}
){ARRAYEXP
:array_typeid 1014
:array_collid 100
:element_typeid 10
42
:elements (
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 12 [ 4
8 0 0 0 49 45 85 8
2 71 69 78 84 ]
}
){CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 4
0 0 0 0 50 45 72 7
3 71 72 ]
}
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23

```

```

:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0
0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey"
"ps_suppkey" "ps_availqt
y" "ps_supplycost" "ps_c
omment")
}
:rtekind 0
:relid 17409
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
}
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
}
:varlevelsup 1
:varreturningtype
0
}
}
}
}

```

```

:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 2
:location -1
}
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 1
:varreturningtype
0
:varnosyn 1
:varattnosyn 3
:location -1
}
)
:location -1
}
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0
0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0
0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname 1_orderkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname 1_supkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname 1_linenum
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname 1_discount

```

```

:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```



```

        :varnosyn 1
        :varattnosyn 14
        :location -1
    }
    :resno 14
    :resname 1_shipinstruct
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 1042
    :vartypmod 14
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location -1
    }
    :resno 15
    :resname 1_shipmode
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 1043
    :vartypmod 48
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location -1
    }
    :resno 16
    :resname 1_comment
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 16
    :resjunk false
    }
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :security_barrier false
    :relid 19014
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    )
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 19014
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 9 10 12 18)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    }
    :quals
    {BOOLEXP
    :boolop and
    :args (
    {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 945
    }
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 957
    }
    )
    }
    :location 955
    }
    )

```

```

{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location 1012
  }
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 1024
  }
)
:location 1022
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 11
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 11
    :location 1079
  }
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 1098
    :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
  }
)
:location 1090
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 11
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 11
    :location 1155
  }
)
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 1173
    :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
  }
  {CONST
    :consttype 1186
    :consttypmod 327679
    :constcollid 0
    :constlen 16
    :constbyval false
    :constisnull false
    :location 1197
    :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ]
  }
)
:location 1186
}
:location 1166
}
:location 1008
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {OPEXPR
      :opno 1760
      :opfuncid 1726
      :opresulttype 1700
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {CONST

```

```

:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location 752
:constvalue 8 [ 32 0 0 0 -1 -128 -120 1
9 ]
}
{AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (0 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 762
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 758
}
}
:location 756
}
:resno 1
:resname ?column?
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 671
}
}
:location 669
}
)
:location 653
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 198
}
:resno 1
:resname ps_supkey
:ressortgrouppref 0
:resorigtbl 19010
:resorigcol 2
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 146
}

```

```

{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 1254
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1268
}
)
:location 1266
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1292
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1301
:constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]
}
)
:location 1299
}
)
:location 1250
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51
}
:resno 1
:resname s_name
:ressortgroupref 1
:resorigtbl 19806
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 67
}
:resno 2
:resname s_address
:ressortgroupref 0
:resorigtbl 19806
:resorigcol 3
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0

```

```

:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs true
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTF false
  :isReturn false
  :ctelist <>
  :rtable (
    {RANGETBENTRY
      :alias <>
      :eref
        {ALIAS
          :aliasname lineitem_secure
          :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
        }
      :rtekind 1
      :subquery <>
      :security_barrier false
      :relid 19014
      :inh false
      :relkind v
      :rellockmode 1
      :perminfoindex 1
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias
        {ALIAS
          :aliasname 1
          :colnames <>
        }
      :eref
        {ALIAS
          :aliasname 1
          :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
        }
      :rtekind 0
      :relid 17440
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 2
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias
        {ALIAS
          :aliasname o
          :colnames <>
        }
      :eref
        {ALIAS
          :aliasname o
          :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
        }
      :rtekind 0
      :relid 17428
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 3
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias
        {ALIAS
          :aliasname ps
          :colnames <>
        }
      :eref
        {ALIAS
          :aliasname ps
          :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
        }
      :rtekind 0
      :relid 17409
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 4
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
  )
}

:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 19014
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 9 10 12 18)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17440
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17428

```

```

:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 2
}
)
:quals <>
}
):rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:varatype 23
:varitypm 0 -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{VAR
:varno 2
:varattno 1
:varatype 23
:varitypm 0 -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 6
:varatype 1042
:varitypm 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0
1 0 0 0 48 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 7
3 71 72 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 4
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 1
:varatype 23
:varitypm 0 -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
}

```

```

{VAR
:varno 2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
)
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location 955
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
{PARAM
:paramkind 1

```

```

:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location 1022
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 11
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1098
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
}
)
:location 1090
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 11
:location -1
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]
}
)
:location 1166
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{OPEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location 752
:constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ]
}
}
{AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtransype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location -1
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 758
}
}
}

```



```

)
:location 756
)
:resno 1
:resname ?column?
:ressortgroupref 0
:resrightbl 0
:resorigcol 0
:resjunk false
)
)
--override 0
--conflist <
--returningOldAlias <
--returningNewAlias <
--returningList <
--groupClause <
--groupDistinct false
--groupingSet <
--havingQual <
--windowClause <
--distinctClause <
--sortClause <
--limitOffset <
--limitCount <
--limitOption <
--rowMarks <
--setOperations <
--constraintDeps <
--withCheckOptions <
--stmt_location <
--stmt_len 0
)

WARNING: is_parallel_safe: INPUT qual = ((RESTRICINFO :clause (OPEXPR :opno 1098 :ofuncid 1098 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 11 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 11 :location -1) (CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause < :rinfo_serial 8 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies < :left_em < :right_em < :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 2345 :ofuncid 2338 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 11 :location -1) (CONST :consttype 1114 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 -49 -115 -1 -1 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant ant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause < :rinfo_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies < :left_em < :right_em < :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1)) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause < :rinfo_serial 11 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paratype 23 :paratypmod -1 :paramcollid 0 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3 4)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvccfreq -1 :right_mvccfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :var
```

```
WARNING: is_parallel_safe: INPUT qual = ((OEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args ((CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ])) {AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 2281 :aggargtypes (o 1700) :aggdirectargs <> :args ((TARGETENTRY :expr {VAR :varno 2 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false)) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1)) :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((OEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args ((CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ])) {AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 2281 :aggargtypes (o 1700) :aggdirectargs <> :args ((TARGETENTRY :expr {VAR :varno 2 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn n 5 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false)) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1)) :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTF false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17006
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17376
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname ANY_subquery
:colnames <>
}
:eref
{ALIAS
:aliasname ANY_subquery
:colnames ("ps_supkey")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 17399
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17376
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
```

```

        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
    }
}
:rtekind 0
:relid 17382
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp_secure
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 19010
:inh false
:relkind v
:rellockmode 1
:perminfoindex 6
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname ANY_subquery
:colnames <>
}
:eref
{ALIAS
:aliasname ANY_subquery
:colnames ("p_partkey")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17409
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17399
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
:rtekind 0
:relid 17387
:inh false
:relkind r
:rellockmode 1
:perminfoindex 9
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 19006
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
}

```

```

:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17382
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 19010
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17387
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 5
}
{RANGETBLREF
:rtindex 6
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1

```

```

        :location -1
    }
    {VAR
    :varno 4
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 4
    :location -1
    }
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 3
:location -1
}
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location -1
}
}
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 6
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 2
}
}
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1268
}
}
:location 1266
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29

```

```

:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1292
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1301
:constvalue 10 [ 40 0 0 0 0 70 82 65 78 67 69 ]
}
)
:location 1299
}
}
}
:rarg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 9
}
)
:quals (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 9
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 3
:location -1
}
)
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:rarg
{RANGETBLREF
:rtindex 10
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 10
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 1
:location -1
}
)
{VAR
:varno 9
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 2
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 10
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 6
:location -1
}
)
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false

```

```

        :constisnull false
        :location -1
        :constvalue 6 [ 24 0 0 0 0 -128 ]
    }
}
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 9
:varattno 3
:vartype 23
:varitypm 1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 3
:location -1
}
)
:location -1
}
{SUBPLAN
:sublinkType 4
:testexpr <>
:params <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 1700
:firstColitypm 1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe false
:setParam <>
:parParam (i 0 1)
:args (
{VAR
:varno 9
:varattno 1
:vartype 23
:varitypm 1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 1
:location -1
}
)
{VAR
:varno 9
:varattno 2
:vartype 23
:varitypm 1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 42.24999999999999
}
)
:location 669
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGEtblref
:rtindex 11
}
)
)
:quals (
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 11
:varattno 2
:vartype 1043
:varitypm 59
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 2
:location 581
}
}
:resulttype 25
:resultitypm 1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:constitypm 1
:constcollid 100
:constlen 1
:constbyval false
:constisnull false
:location 593
:constvalue 11 [ 44 0 0 0 37 105 118 111 114 121 37 ]
}
)
:location 588
}
)
}

```

```

:usingClause <>
:join_using_alias <>
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 9
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 9
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 11
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 1
        :location 411
      }
    )
  }
  :location 327
)
:alias <>
:rtindex 0
}
:usingClause <>
:join_using_alias <>
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 4
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 9
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 9
        :varattnosyn 2
        :location -1
      }
    )
  }
  :location 146
)
:alias <>
:rtindex 0
}
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 4
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
      }
    }
  :resno 1
  :resname s_name
  :ressortgroupref 1
  :resorigtbl 19006
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 4
        :varattno 3
        :vartype 1043
        :vartypmod 44
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 3
        :location -1
      }
    }
  :resno 2
  :resname s_address
  :ressortgroupref 0
  :resorigtbl 19006
  :resorigcol 3
  :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```


[illegible]


```
Sort (cost=63183.47..63183.51 rows=13 width=52) (actual time=160.397..160.402 rows=55.00 loops=1)
  Output: s.s_name, s.s_address
  Sort Key: s.s_name
  Sort Method: quicksort  Memory: 28kB
  Buffers: shared hit=28112
-> Nested Loop Semi Join (cost=979.63..63183.23 rows=13 width=52) (actual time=29.941..159.820 rows=55.00 loops=1)
  Output: s.s_name, s.s_address
  Buffers: shared hit=28109
  -> Nested Loop (cost=4.02..20.80 rows=14 width=56) (actual time=0.177..0.437 rows=69.00 loops=1)
  Output: s.s_name, s.s_address, s.s_supkey
  Buffers: shared hit=42
  -> Nested Loop (cost=2.48..3.91 rows=1 width=8) (actual time=0.108..0.121 rows=1.00 loops=1)
  Output: nation.n_nationkey, n.n_nationkey
  Join Filter: (nation.n_nationkey = n.n_nationkey)
  Rows Removed by Join Filter: 4
  Buffers: shared hit=3
  -> Seq Scan on public.nation (cost=0.00..1.31 rows=1 width=4) (actual time=0.037..0.040 rows=1.00 loops=1)
  Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
  Filter: (nation.n_name = 'FRANCE'::bpchar)
  Rows Removed by Filter: 24
  Buffers: shared hit=1
  -> HashAggregate (cost=2.48..2.53 rows=5 width=4) (actual time=0.067..0.069 rows=5.00 loops=1)
  Output: n.n_nationkey
  Group Key: n.n_nationkey
  Batches: 1  Memory Usage: 32kB
  Buffers: shared hit=2
  -> Hash Join (cost=1.07..2.47 rows=5 width=4) (actual time=0.053..0.061 rows=5.00 loops=1)
  Output: n.n_nationkey
  Hash Cond: (n.n_regionkey = r.r_regionkey)
  Buffers: shared hit=2
  -> Seq Scan on public.nation n (cost=0.00..1.25 rows=25 width=8) (actual time=0.006..0.009 rows=25.00 loops=1)
  Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
  Buffers: shared hit=1
  -> Hash (cost=1.06..1.06 rows=1 width=4) (actual time=0.031..0.031 rows=1.00 loops=1)
  Output: r.r_regionkey
  Buckets: 1024  Batches: 1  Memory Usage: 9kB
  Buffers: shared hit=1
  -> Seq Scan on public.region r (cost=0.00..1.06 rows=1 width=4) (actual time=0.021..0.021 rows=1.00 loops=1)
  Output: r.r_regionkey
  Filter: (r.r_name = 'EUROPE'::bpchar)
  Rows Removed by Filter: 4
  Buffers: shared hit=1
-> Bitmap Heap Scan on public.supplier s (cost=1.54..16.18 rows=72 width=60) (actual time=0.067..0.306 rows=69.00 loops=1)
  Output: s.s_supkey, s.s_name, s.s_address, s.s_nationkey, s.s_phone, s.s_acctbal, s.s_comment
  Recheck Cond: (s.s_nationkey = n.n_nationkey)
  Filter: (s.s_acctbal > '0'::numeric)
  Rows Removed by Filter: 9
  Heap Blocks: exact=37
  Buffers: shared hit=39
  -> Bitmap Index Scan on idx_supplier_nationkey (cost=0.00..1.52 rows=80 width=0) (actual time=0.039..0.039 rows=78.00 loops=1)
  Index Cond: (s.s_nationkey = n.n_nationkey)
  Index Searches: 1
  Buffers: shared hit=2
-> Nested Loop Semi Join (cost=975.61..4505.88 rows=1 width=8) (actual time=2.310..2.310 rows=0.80 loops=69)
  Output: ps.ps_supkey, s_1.s_supkey
  Join Filter: ((s.s_supkey = s_1.s_supkey) AND (ps.ps_supkey = s_1.s_supkey))
  Rows Removed by Join Filter: 741
  Buffers: shared hit=28067
  -> Hash Semi Join (cost=975.61..4365.86 rows=1 width=4) (actual time=2.145..2.145 rows=0.80 loops=69)
  Output: ps.ps_supkey
  Hash Cond: (ps.ps_partkey = part.p_partkey)
  Buffers: shared hit=26700
  -> Index Scan using idx_partsupp_supkey on public.partsupp ps (cost=0.29..3390.47 rows=26 width=8) (actual time=0.178..2.021 rows=13.86 loops=69)
  Output: ps.ps_partkey, ps.ps_supkey, ps.ps_availqty, ps.ps_supplycost, ps.ps_comment
  Index Cond: (ps.ps_supkey = s.s_supkey)
  Filter: ((ps.ps_availqty > 100) AND ((ps.ps_availqty)::numeric > (SubPlan 1)))
  Rows Removed by Filter: 24
  Index Searches: 69
  Buffers: shared hit=25968
  SubPlan 1
  -> Aggregate (cost=42.24..42.25 rows=1 width=32) (actual time=0.049..0.049 rows=1.00 loops=2603)
  Output: (0.5 * sum(1.l_quantity))
  Buffers: shared hit=23196
  -> Nested Loop Semi Join (cost=1.27..42.23 rows=1 width=5) (actual time=0.044..0.048 rows=0.46 loops=2603)
  Output: 1.l_quantity
  Buffers: shared hit=23196
  -> Nested Loop Semi Join (cost=0.85..33.78 rows=1 width=13) (actual time=0.038..0.042 rows=0.46 loops=2603)
  Output: 1.l_quantity, 1.l_partkey, 1.l_supkey
  Buffers: shared hit=19628
  -> Index Scan using idx_lineitem_partkey_supkey_shipdate on public.lineitem l (cost=0.43..8.45 rows=1 width=17) (actual time=0.021..0.023 rows=1.13 loop
s=2603)
  Output: 1.l_orderkey, 1.l_partkey, 1.l_supkey, 1.l_linenum, 1.l_quantity, 1.l_extendedprice, 1.l_discount, 1.l_tax, 1.l_returnflag, 1.l_linestatu
s, 1.l_shipdate, 1.l_commitdate, 1.l_receiptdate, 1.l_shipinstruct, 1.l_shipmode, 1.l_comment
  Index Cond: ((1.l_partkey = ps.ps_partkey) AND (1.l_supkey = ps.ps_supkey) AND (1.l_shipdate >= '1995-01-01'::date) AND (1.l_shipdate < '1996-01-01
00:00:00'::timestamp without time zone))
  Index Searches: 2603
  Buffers: shared hit=10774
  -> Index Only Scan using idx_orders_orderkey_priority on public.orders o (cost=0.42..12.88 rows=1 width=4) (actual time=0.016..0.016 rows=0.40 loops=2951)
  Output: o.o_orderkey, o.o_orderpriority
  Index Cond: ((o.o_orderkey = 1.l_orderkey) AND (o.o_orderpriority = ANY (('{1-URGENT,2-HIGH}'::bpchar))))
  Heap Fetches: 0
  Index Searches: 2951
  Buffers: shared hit=8854
  -> Index Only Scan using idx_partsupp_partkey_supkey_availqty on public.partsupp ps_1 (cost=0.42..8.44 rows=1 width=8) (actual time=0.014..0.014 rows=1.00 loo
ps=1189)
  Output: ps_1.ps_partkey, ps_1.ps_supkey, ps_1.ps_availqty
  Index Cond: ((ps_1.ps_partkey = ps.ps_partkey) AND (ps_1.ps_supkey = ps.ps_supkey) AND (ps_1.ps_availqty > 0))
  Heap Fetches: 0
  Index Searches: 1189
  Buffers: shared hit=3568
-> Hash (cost=960.17..960.17 rows=1212 width=4) (actual time=8.239..8.240 rows=2082.00 loops=1)
  Output: part.p_partkey
  Buckets: 4096 (originally 2048)  Batches: 1 (originally 1)  Memory Usage: 106kB
  Buffers: shared hit=762
  -> Bitmap Heap Scan on public.part (cost=36.13..960.17 rows=1212 width=4) (actual time=1.777..7.680 rows=2082.00 loops=1)
  Output: part.p_partkey
  Recheck Cond: ((part.p_name)::text ~ '%ivory% '::text)
  Heap Blocks: exact=752
  Buffers: shared hit=762
  -> Bitmap Index Scan on idx_part_name_trgm (cost=0.00..35.83 rows=1212 width=0) (actual time=1.639..1.639 rows=2082.00 loops=1)
  Index Cond: ((part.p_name)::text ~ '%ivory% '::text)
  Index Searches: 1
  Buffers: shared hit=10
-> Seq Scan on public.supplier s_1 (cost=0.00..113.00 rows=1801 width=4) (actual time=0.002..0.166 rows=930.85 loops=55)
  Output: s_1.s_supkey, s_1.s_name, s_1.s_address, s_1.s_nationkey, s_1.s_phone, s_1.s_acctbal, s_1.s_comment
  Filter: (s_1.s_acctbal > '0'::numeric)
  Rows Removed by Filter: 98
  Buffers: shared hit=1337

Planning:
  Buffers: shared hit=472
  Planning Time: 71.829 ms
  Execution Time: 162.409 ms
(118 rows)

tpch-200mb=>
```